

Performance Testing

Date	01 July 2026
Team ID	N/A
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Complete Streamlit Application 10 120 Application remains responsive without failures

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	PyTest, httpx TestClient
Type of Testing	Performance Testing, API Testing, Load Testing
Target Module / API	Event Analyzer, Topic Generator, Fact Checker, FastAPI Endpoints
Test Environment	Localhost (FastAPI + Streamlit)
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Event Analysis API	10	60	Returns event categories within target response time
2	Topic Generation API	10	60	Generates networking topics successfully
3	Fact Checker API	10	60	Returns verified topic summaries
4	Complete Streamlit Application	10	120	Application remains responsive without failures

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	>20 req / sec	2.4 req / sec	Pass	Stable throughput
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	48%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	55%	Pass	Memory usage within limits

Step 4: Observations & Analysis

Key Findings:

The Personalized Networking Assistant performed efficiently under normal load. All AI services responded correctly, and FastAPI endpoints maintained stable performance throughout testing.

Bottlenecks Identified:

Minor delay was observed during the first AI model initialization. Subsequent requests were processed quickly with no significant bottlenecks.

Optimization Steps Taken:

- Modularized backend services.
- Optimized API routing.
- Reused loaded AI models.
- Improved request handling using FastAPI.
- Reduced unnecessary computations for faster responses.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshot 1:

Personalized Networking Assistant 1.0.0 OAS 3.1

/openapi.json

default ^		
GET	/health	Health Check
POST	/analyze-event	Analyze Event Route
POST	/generate-conversation	Generate Conversation Route
POST	/fact-check	Fact Check Route
POST	/feedback	Feedback Route
GET	/history	History Route
GET	/feedback-history	Feedback History Route
GET	/	Home

Screenshot 2:

Personalized Networking Assistant

Generate smart conversation starters for networking events.

Enter Event Description

AI & Tech Innovation Summit 2026 focused on Artificial Intelligence, Machine Learning, Data Science, Cloud Computing, Cybersecurity, and Startup Innovation. The event includes keynote sessions, workshops, networking opportunities, and startup showcases.

Enter your interests separated by commas

Artificial Intelligence, Machine Learning, Python, Data Science, Web Development, Startups

Generate Conversation Starters

Extracted Themes

```
[
  0 : "Artificial Intelligence"
  1 : "Cloud Computing"
  2 : "Data Science"
]
```

Conversation Starters

• I noticed you're interested in Artificial Intelligence. What projects have you been working on?

HelpfulNot Helpful

• I noticed you're interested in Cloud Computing. What projects have you been working on?

HelpfulNot Helpful

• I noticed you're interested in Data Science. What projects have you been working on?

HelpfulNot Helpful